

- 14** In a heating experiment, energy is supplied at a constant rate to a liquid in a beaker of negligible heat capacity. The temperature of the liquid rises by 4.0 K per minute just before it begins to boil. After 40 minutes, all the liquid has boiled away.

For this liquid, what is the ratio $\frac{\text{specific heat capacity}}{\text{specific latent heat of vapourisation}}$?

A $\frac{1}{10} \text{ K}^{-1}$

B $\frac{1}{40} \text{ K}^{-1}$

C $\frac{1}{160} \text{ K}^{-1}$

D $\frac{1}{640} \text{ K}^{-1}$

$$P = m c \, d\theta/dt = dm/dt \, L$$

$$c / L = (m/40) / (m \times 4.0) = 1/160$$

- 15** The compressions and rarefactions of a sound wave travelling in air in the positive x direction